of Dubai's T3, which is spread over a massive 18.4 mn sq. ft. Changi's four terminals have a combined floor area of 13.6 mn sq. ft, the largest being T3 with 4.1 mn sq. ft. IGI's T1 has now expanded to 169,000 sq. metres from the existing 60,000 sq. metres, since Delhi International Airport Ltd (DIAL) intends to gradually shift its entire domestic operations here, thereby freeing up capacity in T3 solely for international operations. Long-term plans include adding a fourth terminal too.

Besides its four runways, Delhi has added a 3,000 sq. m. I-I transfer area, double the size of the original. "In the first phase, the focus is on increasing the share of domestic to international passengers who choose Indian carriers for direct connectivity," says Videh Kumar Jaipuriar, CEO of DIAL. "For that, we are increasing our flight handling capacity and working with the airlines to create slot banks." A 'slot bank' ensures efficient flight scheduling. For instance, if an airport has North

IMMIGRATION AND SECURITY

Delhi has 103 immigration counters, but it is not uncommon to see long, slow-moving queues of passengers at these counters, especially during peak departure and arrival hours. In global hubs, large resources are invested into this critical process. Dubai, for instance, has 107 immigration counters at just the arrivals terminal and 60 automated smart gates in Terminals 1 and 3, which together serve 95 per cent of its international passengers. At its departures, the two terminals combined have 71 counters and 45 smart gates segregated class-wise. Singapore, on the other hand, has plans to process almost 95 per cent of its arrivals through automated lanes.

In India, the civil aviation ministry's latest meeting with airport operators, security agencies and the Bureau of Immigration delved into security solutions derived from studying international airport models such as those of Singapore and



V.K. Jaipuriar
CEO, DIAL

"IN PHASE 1, WE'LL FOCUS ON INCREASING THE SHARE OF DOMESTIC TO INTERNATIONAL PASSENGERS FLYING INDIAN CARRIERS FOR DIRECT CONNECTIVITY. FOR THAT, WE'RE INCREASING OUR FLIGHT HANDLING CAPACITY"

America departures at 1 am, domestic flights should arrive two hours prior, bringing North America-bound passengers. This strategy maximises plane capacity by combining passengers originating from Delhi with those connecting flights. Most hubs worldwide use this system, where flights to a major destination take off in quick succession at designated time slots.

DIAL also needs to set up an airport operations centre (APOC), the planning and management tool to enhance airport operations efficiency and reduce recovery time after disruptions. Large players may also need terminals dedicated to them, like Emirates has in Dubai. Air India already operates from Delhi's T3.

Canada. There were talks of cross-utilising X-ray baggage machines in domestic and international terminals in Delhi to cut wait time, while proof of concept trials for the use of e-gates and e-biometrics for immigration are already under way.

The stakeholders have also highlighted more passenger-friendly security checks. They cited examples like the 'One Schengen' security framework in European airports wherein once passengers check into an airport, they are not subjected to further security checks while in transit at other airports. India is mulling moves like eliminating physical boarding passes for international passengers at security checks. Its novel DigiYatra app, which helps passengers check in simply by using biometrics, might also need to expand to cover international passengers. "There is also scope for granting short-duration visas to international transit passengers," says a government official.

BAGGAGE HANDLING

Global hubs are forever trying to handle more bags safely in the shortest possible time. For instance, the Dubai International Airport, specifically Terminal 3, has the largest baggage handling system globally. It includes 21 screening injection points, 49 make-up carousels and 4,500 Early Baggage Storage (EBS) po-

THE DELHI UPGRADE

An aviation hub facilitates seamless transit connectivity for travel worldwide using a hub-and-spoke approach. To become one, the Delhi IGI airport has to undertake a massive upgrade to compete with world-class hubs such as Singapore and Dubai

THE SIX ESSENTIALS



1. IMMIGRATION

NEED FOR AUTOMATED SMART GATES

DELHI: 103 counters

DUBAI: 107 counters at arrivals terminal, 60 automated smart gates

SINGAPORE: 95% arrivals through automated lanes

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